# National 5 Engineering Science

## Analogue Electronics Worksheet – Mixed Questions

Show all working. Use correct units.

1. A 12V supply is connected to a 240Ω resistor. Calculate the current.

2. Two resistors of 15Ω and 35Ω are connected in series to a 10V supply.
a) Calculate the total resistance.
b) Calculate the circuit current.
c) Calculate the voltage across each resistor.

3. Two resistors of 100Ω and 300Ω are connected in parallel to a 12V supply.
a) Calculate the total resistance.
b) Calculate the total circuit current.
c) Calculate the current through the 100Ω resistor.

4. A current of 0.5A flows through a resistor connected to a 9V supply. Calculate the resistance.

5. A 20Ω resistor is connected in series with two resistors (60Ω and 30Ω) in parallel. The supply voltage is 18V.
a) Calculate the total resistance.
b) Calculate the circuit current.
c) Calculate the voltage across the 20Ω resistor.

6. R1 = 2.2kΩ, R2 = 4.7kΩ, Vs = 12V.
a) Calculate total resistance.

b) Calculate circuit current.
c) Calculate the voltage across R2.

7. Three resistors (10Ω, 20Ω, 40Ω) are connected in parallel to a 12V supply.
a) Calculate total resistance.
b) Calculate total current.
c) Calculate current through the 20Ω resistor.

8. A current of 2A flows through a 15Ω resistor. Calculate the supply voltage.

9. R1 = 100Ω, R2 = 150Ω, R3 = 250Ω, Vs = 20V.
a) Calculate total resistance.
b) Calculate circuit current.
c) Calculate the voltage across R3.

10. Two resistors of 8Ω and 12Ω are connected in parallel to a 6V supply.
a) Calculate total resistance.
b) Calculate total current.

11. A 50Ω resistor is connected in series with a parallel pair of 100Ω and 100Ω resistors. The supply voltage is 10V.
a) Calculate total resistance.
b) Calculate circuit current.
c) Calculate current through one of the 100Ω resistors.

12. A student needs a 25Ω resistor but only has two 50Ω resistors.
a) Draw a circuit to achieve 25Ω.
b) Prove your answer with calculations.

13. A 24V supply is connected to a 3kΩ resistor. Calculate the current (in mA).

14. Two resistors (1kΩ and 2kΩ) are connected in parallel to a 9V supply.
a) Calculate total resistance.
b) Calculate total current.

15. Two resistors (30Ω and 60Ω) are connected in parallel. This combination is then connected in series with a 10Ω resistor. Supply voltage is 12V.
a) Total resistance
b) Circuit current
c) Voltage across the 10Ω resistor
d) Current through the 30Ω resistor

16. R1 = 470Ω, R2 = 1.5kΩ, Vs = 9V.
a) Calculate total resistance.
b) Calculate circuit current.

17. Two heaters of resistance 20Ω and 30Ω are connected in parallel to a 12V battery.
a) Calculate total resistance.
b) Calculate total current drawn from the battery.
c) Calculate current through the 20Ω heater.

18. A circuit has a total resistance of 40Ω. One resistor of 10Ω is in series with a parallel combination.
Calculate the resistance of the parallel section.

19. Resistors of 2kΩ, 5kΩ and 10kΩ are connected in parallel to a 10V supply.
a) Calculate total resistance.
b) Calculate total current.
c) Calculate current through the 2kΩ resistor.

20. A 6V supply feeds a circuit where a 15Ω resistor is in series with two 30Ω resistors connected in parallel.
a) Calculate total resistance.
b) Calculate circuit current.
c) Calculate voltage across the parallel section.
d) Calculate current through one of the 30Ω resistors.